Python-Test-03 **21-03-25**

MCQ-10M

1. What is the correct way to create a list in Python?

a) {1, 2, 3}

b) (1, 2, 3)

c) [1, 2, 3]

d) list(1, 2, 3)

2. What will be the output of the following code?

```
list1 = [1, 2, 3, 4, 5]
print(list1[1:4])
```

a) [2, 3, 4]

b) [1, 2, 3, 4]

c) [2, 3, 4, 5]

d) Error

3. Which method is used to remove an item from a list by value?

a) pop()

b) remove()

c) del

d) discard()

4. What does list.append(5) do?

a) Adds 5 to the end of the list

b) Adds 5 to the beginning of the list

c) Replaces all elements with 5

d) Removes 5 from the list

5. What is the key difference between a list and a tuple?

a) Lists are immutable, tuples are mutable

b) Tuples are immutable, lists are mutable

c) Both are immutable

d) Both are mutable

6. How do you create a tuple with a single element?

a) (1)

b) (1,)

c) [1]

d) {1}

7. What will tup = (1, 2, 3); tup[0] = 5 result in?

a) (5, 2, 3)

b) (1, 2, 3, 5)

c) TypeError: 'tuple' object does not support item assignment

d) None

8. Which of the following functions converts a list into a tuple?

a) tuple()

b) list()

c) set()

d) dict()

9. What is the correct way to access the value associated with a key in a dictionary?

a) dict.key

b) dict[key]

c) dict{key}

d) dict.get[key]

10. What will dict1 = {'a': 1, 'b': 2}; print(dict1.get('c', 0)) output?

a) None

b) 0

c) KeyError

d) 'c'

11. Which method removes a key-value pair from a dictionary?

a) delete()

b) pop()

c) discard()

d) remove()

12. What happens if you try to access a key that does not exist in a dictionary using dict[key]?

a) Returns None

b) Raises a KeyError

c) Returns an empty dictionary

d) Creates a new key

13. What is the key characteristic of a Python set?

a) Ordered and allows duplicates

b) Unordered and allows duplicates

c) Ordered and does not allow duplicates

d) Unordered and does not allow duplicates

14. How do you define an empty set?

a) {}

b) set()

c) []

d) ()

15. Which of the following operations can be performed on sets?

a) Union

b) Intersection

c) Difference

d) All of the above

16. What will be the output of the following?

set1 = {1, 2, 3}

set1.add(4)

print(set1)

a) {1, 2, 3, 4}

b) [1, 2, 3, 4]

c) (1, 2, 3, 4)

d) Error

17. How do you define a function in Python?

a) def function_name():

b) function function_name():

c) define function_name():

d) func function_name():

18. What will return do inside a function?

a) Stops the function and returns a value

b) Continues execution

c) Restarts the function

d) Prints the value and stops execution

19. What will be the output of the following function call?

```
def test(a, b=2, c=3):
    return a + b + c


print(test(1, c=4))
```

a) 6

b) 7

c) 8

d) 9

20. What is the key difference between a list and a tuple in Python?

a) Lists are immutable, while tuples are mutable.

b) Tuples are immutable, while lists are mutable.

c) Both lists and tuples are immutable.

d) Both lists and tuples are mutable.

Section 1: List (10 Marks)

Q1. List Manipulation (5 Marks)

Write a Python program to:

Take a list of numbers as input.

Remove duplicates while maintaining order.

Sort the remaining elements in descending order.

Return the second largest element.

Example:

Input: [4, 7, 2, 9, 7, 1, 9, 2]

Output: 7

Q2. Rotate a List (5 Marks)

Write a function rotate_list(lst, k) that rotates the list by k positions to the right.

Example:

Input: lst = [1, 2, 3, 4, 5], k = 2

Output: [4, 5, 1, 2, 3]

Section 2: Tuple (5 Marks)

Q3. Tuple Operations (5 Marks)

Write a Python function to:

Convert a list of numbers to a tuple.

Find the maximum and minimum numbers in the tuple.

Return them as a tuple.

Example:

Input: [10, 3, 5, 8, 2]

Output: (10, 2)

Section 3: Set (5 Marks)

Q4. Set Operations (5 Marks)

Write a Python function that:

Takes two sets as input.

Returns their union, intersection, and difference.

Input: set1 = {1, 2, 3}, set2 = {3, 4, 5}

Output: ({1, 2, 3, 4, 5}, {3}, {1, 2})

Section 4: Dictionary (10 Marks)

Q5. Frequency Count (5 Marks)

Write a Python function count_frequency(lst) that counts the occurrences of each element in a list and stores the results in a dictionary.

Example:

Input: [1, 2, 2, 3, 3, 3, 4, 4, 4, 4]

Output: {1: 1, 2: 2, 3: 3, 4: 4}

Q6. Invert a Dictionary (5 Marks)

Write a function invert_dict(d) that swaps keys and values in a dictionary.

Example:

Input: {'a': 1, 'b': 2, 'c': 3}

Output: {1: 'a', 2: 'b', 3: 'c'}

Section 5: Functions (10 Marks)

Q7. Palindrome Checker (5 Marks)

Write a function is_palindrome(s) that checks if a given string is a palindrome (ignoring case and spaces).

Example:

Input: "Madam"

Output: True

Q8. Fibonacci Series (5 Marks)

Write a function fibonacci(n) that generates the first n terms of the Fibonacci sequence.

Example:

Input: n = 6

Output: [0, 1, 1, 2, 3, 5]